

Radio Resource Management for C-V2X Using Graph Matching and Actor–Critic Learning

Qiuzhan Zhou^{ID}, Chi Guo^{ID}, Cong Wang^{ID}, *Member, IEEE*, and Lin Cui^{ID}

Abstract—We propose a hybrid centralized-distributed radio resource management (RRM) scheme for cellular vehicle-to-everything (C-V2X), which is to mitigate the interference caused by radio resource sharing between vehicle-to-infrastructure (V2I) links and vehicle-to-vehicle (V2V) links. Specifically, it contains two decoupled aspects: centralized channel allocation and distributed power control. The former is conducted by graph matching to decide spectrum sharing strategy; the latter is performed via actor-critic learning, a deep deterministic policy gradient (DDPG)-based algorithm, to increase the total system capacity. Finally, the proposed scheme is numerically evaluated and outperforms other deep Q-network (DQN)-based schemes in terms of the system capacity.

Index Terms—C-V2X, radio resource management, graph matching, actor-critic learning.

I. INTRODUCTION

THE TERM vehicle-to-everything (V2X) refers to the vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-pedestrian (V2P), vehicle-to-network (V2N) [1]. As one of the key enablers for intelligent transportation system (ITS), cellular vehicle-to-everything (C-V2X) is proposed in Release 14 to improve the spectrum utilization efficiency for ITS [2]. The basic idea is that device-to-device (D2D)-based V2V links and V2I links are permitted to access the same cellular radio resources. It is worth noting that V2V links are intended for safety-critical services that require high-reliability. In this case, the spectrum sharing between V2I links and V2V links should guarantee the reliability of V2V links. Therefore, radio resource management (RRM) is motivated and aims to devise spectrum sharing.

RRM contains two coupled aspects: channel allocation and power control. Specifically, channel allocation is performed to decide which V2I link to share spectrum with V2V links. Power control is conducted to mitigate interference over sharing radio channels. Both aspects depend on accurate channel state information (CSI) of all links, which is a challenge due to the high mobility of vehicle [3]. Based on how CSI is utilized, RRM can be done in either a centralized or distributed manner.

Centralized RRM, requires that the base station (BS) collects CSI of all links, where CSI of V2V links is inaccurate and time-consuming [4]. For instance, some works have to rely on inaccurate CSI estimation or feedback, incurring a lot of

signaling overload [5], [6]. Therefore, the applicability of centralized RRM is limited. Unlike centralized RRM, distributed RRM is executed by every vehicular user equipment (V-UE) that interacts with each other to obtain local CSI [7]. Although distributed RRM has decent immunity to high mobility, it typically relies on the known channel allocation.

Having seen the limitations of both centralized and distributed RRM, we propose a novel hybrid centralized-distributed RRM scheme inspired by [8], [9]. Specifically, channel allocation is performed by the BS while power control is implemented among V-UEs. Unlike [8], [9], our scheme not only considers the practical implementation but also focuses on the optimization formulation of RRM, a mixed-integer programming. Specifically, the optimization variable for channel allocation is discrete and that for power control is continuous, [10]. Many works, with deep Q-network (DQN) algorithms, apply quantization to continuous variable, lowering the system capacity [11], [12]. To avoid this issue, we decouple two aspects so that they can be independently handled by different methods.

In this letter, our goal is to maximize the system capacity while guaranteeing the reliability of V2V links. The key contributions are summarized as follows:

- We decouple a RRM problem into two sub-problems: channel allocation and power control. It enables discrete optimization variable and continuous optimization variable to be independently handled in different methods, which offers a lower bound for the system capacity.
- We propose a hybrid centralized-distributed RRM scheme, wherein the BS centrally coordinates channel allocation based on graph matching model, while V-UEs independently perform deep deterministic policy gradient (DDPG)-based algorithm for power control.
- To decrease the computing complexity of DDPG, we apply centralized training and distributed execution. Here, agents can execute power control in distributed manner.

The remainder of this letter is organized as follows. In Section II, we establish the system model and formulate the RRM problem. In Section III, the proposed RRM scheme is elaborated, including centralized channel allocation and distributed power control. Simulation results are discussed in Section IV.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

We consider a single-cell scenario in Fig. 1 with M V2I links and N V2V links, which are indexed by $m \in \mathcal{M} = \{1, \dots, M\}$, $n \in \mathcal{N} = \{1, \dots, N\}$, respectively. It is worth noting that every vehicle establishes V2V links with neighboring vehicles, which is intended for safety-critical services. Meanwhile, each vehicle transmits in V2I mode for non-safety services, enhancing the driving experience [4].

In this scenario, we assume that M V2I links are preassigned to M orthogonal channels (i.e., resource block (RB)) with fixed BS transmission power, i.e., the m -th V2I link occupies the m -th RB. Moreover, every V2I link is occupied by exactly one V2V link, and every V2V link is shared with a V2I link. In

Manuscript received 2 June 2022; revised 14 August 2022; accepted 3 October 2022. Date of publication 10 October 2022; date of current version 9 December 2022. This work was supported in part by the Exploration Foundation of State Key Laboratory of Automotive Simulation and Control, Jilin University, under Grant ascl-zytzxm-202016, and in part by the National Nature Science Foundation of China under Grant 62101210. The associate editor coordinating the review of this article and approving it for publication was F. Tariq. (Corresponding author: Cong Wang.)

The authors are with the College of Communication Engineering, Jilin University, Changchun 130022, China (e-mail: zhouqz@jlu.edu.cn; guochi19@mails.jlu.edu.cn; wangcong2020@jlu.edu.cn; cuilin19@mails.jlu.edu.cn).

Digital Object Identifier 10.1109/LWC.2022.3213176

2162-2345 © 2022 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

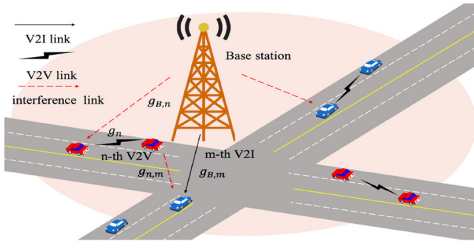


Fig. 1. V2X links in a single-cell scenario.

this case, the signal-to-interference-plus-noise ratio (SINR) of the m -th V2I link and n -th V2V link are given as:

$$\gamma_m^c = \frac{S \cdot g_{B,m}}{\sigma^2 + \sum_n \rho_{n,m} P_n^d g_{n,m}}, \quad (1)$$

$$\gamma_n^v = \frac{P_n^d g_n}{\sigma^2 + S \cdot g_{B,n}}, \quad (2)$$

where S denotes the BS transmission power, and P_n^d is transmission power variable of the n -th V2V link. $g_{B,m}$ and $g_{B,n}$ represent the channel gain from the BS to the m -th V2I link and n -th V2V link, respectively. g_n and $g_{n,m}$ correspond to the n -th V2V link channel gain and interference gain from the n -th V2V link to the m -th V2I link. σ^2 is the noise power. $\rho_{n,m}$ is a binary channel allocation variable, $\rho_{n,m} = 1$ if the m -th V2I link shares spectrum with the n -th V2V link, otherwise $\rho_{n,m} = 0$.

Derived from SINR, the capacity of the m -th V2I link is defined as (3), where W is the bandwidth of each RB.

$$C_m^c = W \log(1 + \gamma_m^c). \quad (3)$$

Similarly, the capacity of the n -th V2V link is given by:

$$C_n^v = W \log(1 + \gamma_n^v). \quad (4)$$

B. Problem Formulation

Based on the above definitions, the RRM can be mathematically formulated as (5) (i.e., **RRM**). Our objective is to maximize the system capacity including all V2I links and V2V links while guaranteeing the reliability of V2V links.

$$[\mathbf{RRM}] \quad \max_{\rho_{n,m}, P_n^d} \left(\sum_m C_m^c(\rho_{n,m}, \mathcal{P}) + \sum_n C_n^v(P_n^d) \right) \quad (5)$$

$$\text{s.t.} \quad \sum_n \rho_{n,m} = 1, \quad \forall m \quad (5a)$$

$$\sum_m \rho_{n,m} = 1, \quad \forall n \quad (5b)$$

$$\rho_{n,m} \in \{0, 1\}, \quad \forall n, m \quad (5c)$$

$$0 \leq P_n^d \leq P_{max}^d, \quad \forall n \quad (5d)$$

$$\gamma_n^v \geq \gamma_{n,min}, \quad \forall n \quad (5e)$$

where $\mathcal{P} = \{P_1^d, \dots, P_n^d, \dots, P_N^d\}$ is a set for P_n^d , P_{max}^d is the maximum transmission power allowed for all the V-UEs, and $\gamma_{n,min}$ is the minimum SINR needed to establish a reliable V2V link. To avoid the mutual interference between multiple V2V links and multiple V2I links, the matching relationship between them is one-to-one matching. Moreover, constraint (5a) ensures unique RB assignment for V2V links, (5b) guarantees the orthogonal RB allocation among V2I links. Constraint (5d) bounds transmission power for every V-UE

and (5e) enforces the minimum SINR requirement of every safety-critical V2V link, which guarantees the reliability of it.

Proposition 1: **RRM** can be decomposed into the two sub-problems, i.e., channel allocation (**CA**) and power control (**PC**), which provides a lower bound for the objective function of **RRM**. Here, the optimal solution to **CA** is denoted by $\rho_{n,m}^{**}$. $C_m^c(\rho_{n,m}) \triangleq C_m^c(\rho_{n,m}, \mathcal{P} = \{P_{max}^d, \dots\})$.

$$[\mathbf{CA}] \quad \max_{\rho_{n,m}} \sum_m C_m^c(\rho_{n,m}) \quad (6)$$

$$\text{s.t.} : (5a), (5b), (5c).$$

$$[\mathbf{PC}] \quad \max_{P_n^d} \left(\sum_m C_m^c(\rho_{n,m}^{**}, \mathcal{P}) + \sum_n C_n^v(P_n^d) \right) \quad (7)$$

$$\text{s.t.} : (5d), (5e).$$

Proof: The optimal solution to **RRM** and **PC** are denoted by $\rho_{n,m}^*$, P_n^{d*} , $\mathcal{P}^* = \{P_n^{d*} \dots\}$ and P_n^{d**} , $\mathcal{P}^{**} = \{P_n^{d**} \dots\}$, respectively. First, let us consider the maximum interference power from V2V links to V2I links. That means power variable P_n^d is replaced by the constant P_{max}^d , i.e.,

$$\begin{aligned} & \sum_m C_m^c(\rho_{n,m}, \mathcal{P}) \\ & \geq \sum_m W \log \left(1 + \frac{S \cdot g_{B,m}}{\sigma^2 + \sum_n \rho_{n,m} P_{max}^d g_{n,m}} \right) = \sum_m C_m^c(\rho_{n,m}). \end{aligned} \quad (8)$$

The above inequality provides a lower bound for V2I links' capacity, which is the objective function of **CA** (determined via only $\rho_{n,m}$). Therefore, we can obtain the local optimum $\sum_m C_m^c(\rho_{n,m}^{**})$ by solving **CA** independently, which achieves the decoupling of $\rho_{n,m}$ and P_n^d . Based on the optimal solution $\rho_{n,m}^{**}$, **RRM** is transformed into **PC** (determined via only P_n^d). Because $\rho_{n,m}^{**}$, P_n^{d**} are included in the feasible set of **RRM** and $\rho_{n,m}^*$, P_n^{d*} are optimizers of **RRM**, we have:

$$\begin{aligned} & \sum_m C_m^c(\rho_{n,m}^*, \mathcal{P}^*) + \sum_n C_n^v(P_n^{d*}) \\ & \geq \sum_m C_m^c(\rho_{n,m}^{**}, \mathcal{P}^{**}) + \sum_n C_n^v(P_n^{d**}). \end{aligned} \quad (9)$$

The above inequality provides a lower bound for the optimal solution of **RRM**.

Based on Proposition 1, **RRM** can be decomposed into two independent sub-problems, i.e., **CA** and **PC**. Specifically, **CA** is an integer programming about $\rho_{n,m}$ whose objective function is the capacity of V2I links; **PC**, aimed at improving the total system capacity, is a continuous optimization problem about P_n^d . The detailed scheme will be discussed in Section III.

III. THE PROPOSED RRM SCHEME

The sketch of the proposed RRM scheme is shown in Fig. 2. First, centralized channel allocation is conducted depending on global CSI. In detail, the BS broadcasts reference signal to V-UEs and receives feedback to evaluate its CSI via an intra-cell dedicated control channel. With global CSI, the BS constructs the graph matching model and broadcasts results to every V-UE. Given the channel allocation, the BS is responsible for the centralized training of the DDPG-based algorithm while every V-UE collects local CSI and executes the policy in a distributed manner. Hence, this is mode 3 defined in the C-V2X, where RRM is assisted by the cellular infrastructure.

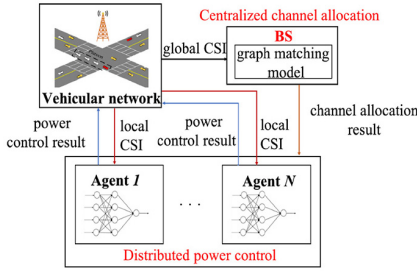


Fig. 2. Diagram of the proposed RRM scheme.

A. Centralized Channel Allocation

The bipartite graph matching is modeled where the vertexes are V2I links $\mathcal{M}$ and V2V links $\mathcal{N}$, respectively. The edge of bipartite graph represents the matching relation $\rho_{n,m}$ between $\mathcal{N}$ and $\mathcal{M}$, which means whether there is a shared channel between n -th V2V link and m -th V2I link or not. After establishing the bipartite graph model, we provide a proposition that shows how CA is equivalent to a maximum weighted matching (MWM) problem. Thus, it can be efficiently solved in polynomial time by Hungarian algorithm.¹

Proposition 2: CA is equivalent to the following MWM problem.

$$\begin{aligned} \max_{\rho_{n,m}} \quad & \sum_m \sum_n W \cdot \rho_{n,m} \cdot w_{n,m} \\ \text{s.t.} \quad & w_{n,m} = -P_{max}^d g_{n,m}, \\ & (5a), (5b), (5c). \end{aligned} \quad (10)$$

Proof: The objective function of CA can be transformed into the following formulation:

$$\begin{aligned} \max_{\rho_{n,m}} \quad & \sum_m W \left(\log(S \cdot g_{B,m}) - \log\left(\sum_n \rho_{n,m} P_{max}^d g_{n,m}\right) \right) \\ \Leftrightarrow \quad & \max_{\rho_{n,m}} \sum_m W \left(-\log\left(\sum_n \rho_{n,m} P_{max}^d g_{n,m}\right) \right) \\ \Leftrightarrow \quad & \max_{\rho_{n,m}} \sum_m \sum_n W \left(-\rho_{n,m} P_{max}^d g_{n,m} \right) \\ \Leftrightarrow \quad & \max_{\rho_{n,m}} \sum_m \sum_n W \cdot \rho_{n,m} \cdot \left(-P_{max}^d g_{n,m} \right) \end{aligned} \quad (11)$$

Through the above transformation, we can obtain an equivalent formulation with CA and achieves the decoupling between channel allocation variables $\rho_{n,m}, \forall n, \forall m$. Specifically, the fixed term $-P_{max}^d g_{n,m}$ can be seen as the weight $w_{n,m}$ corresponding to every edge $\rho_{n,m}$. In addition, constraints (5a), (5b) and (5c) remain unchanged. Hence, CA is converted to find a matching relationship to maximize Equation (11) that fits perfectly into the MWM problem. ■

B. Distributed Power Control

PC is non-convex with continuous power variable, which cannot be solved by the convex methods. In theory, it can be handled by the approximation methods. However, these methods have unacceptable computation complexity and cannot guarantee the reliable performance. Thereby, we apply an actor-critic learning algorithm specialized in continuous action space, DDPG, to cope with it and achieve desired performance.

¹The complexity is upper bounded by $O((\max\{M, N\})^3)$ [13].

Technically, PC is first modeled as a Markov Decision Process (MDP), where every V-UE acts as an agent and interacts with the environment to gain experiences for improving policy.

1) *MDP Framework Formulation:* In the MDP, an agent selects an action with reference to only the current state and gets an instant reward used as feedback to adjust the agent's next action. In this model, three key elements of the MDP, i.e., state space $\mathcal{S}$, action space $\mathcal{A}$, immediate return r_t , are defined as follows:

- **State Space:** Let $s_t^n \in \mathcal{S}$ denotes the current state at time step t of the n -th agent. It is local CSI, including $g_n, g_{B,n}, g_{n,m}, g_{B,m}$. Therein, $g_n, g_{n,m}$ can be obtained through interactions between agents; $g_{B,m}, g_{B,n}$ are detected by the BS. Therein, the central communication cost is relieved by the information exchange between agents.
- **Action Space:** The range for transmission power of the n -th agent, i.e., $\mathbf{a}_t^n = P_t^n \in [0, P_{max}^d]$.
- **Immediate Return:** After $\mathbf{a}_t^n, n \in \mathcal{N}$ is executed, the agents will get a cooperative immediate reward, which is defined as the objective function of PC.

$$r_t = \sum_m C_m^c + \sum_n C_n^v. \quad (12)$$

The goal of agents is to find an optimal policy π , which decides the optimal action $\mathbf{a}^*$ at each state $\mathbf{s}$, i.e., $\mathbf{a}^* = \arg\max_{\mathbf{a}} Q^\pi(\mathbf{s}, \mathbf{a})$. The action-value function $Q^\pi(\mathbf{s}, \mathbf{a})$ is the expected long-term accumulative discounted reward as:

$$Q^\pi(\mathbf{s}, \mathbf{a}) = \mathbb{E}_\pi \left\{ \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \mid s_t = \mathbf{s}, \mathbf{a}_t = \mathbf{a} \right\} \quad (13)$$

where $\gamma \in [0, 1]$ is a discount factor and k is a time step counting symbol. The $Q^\pi(\mathbf{s}, \mathbf{a})$ can be rewritten as:

$$Q^\pi(\mathbf{s}, \mathbf{a}) = \mathbb{E} \{ r_{t+1} + \gamma Q^\pi(s_{t+1}, \mathbf{a}_{t+1}) \mid s_t = \mathbf{s}, \mathbf{a}_t = \mathbf{a} \}. \quad (14)$$

This recursive relationship, called the Bellman equation, can be used to find the optimal policy.

2) *DDPG-Based Algorithm:* Combined with the advantage of policy gradient and target network in DQN, DDPG is an enhanced actor-critic learning algorithm. In order to avoid instability caused by competition among multiple agents and achieve distributed power control, we apply the centralized training and distributed execution to it. Specifically, as shown in Fig. 3, the centralized training is implemented with collected experiences in a replay memory by the BS, while the distributed policy is executed by every agent. Once the centralized training is finished, distributed policy is executed based on local CSI and is updated every T_u episodes. Hence, the above structure not only guarantees the fast convergence of DDPG, but also realizes distributed decision-making for every agent.

In addition, to accelerate the convergence of deep neural network (DNN), DDPG adopts quasi-static target network approach and defines two separate networks in the training process. Specifically, each agent consists of two parts: the actor and the critic. In each part, we use two DNNs to perceive complex environmental states. They are with the same structure but different parameters, i.e., an estimated network updated in real time and a target network with soft updated parameters. The estimated network and the target network in the actor (or critic) are denoted by parameters θ^μ and $\theta^{\mu'}$ (or

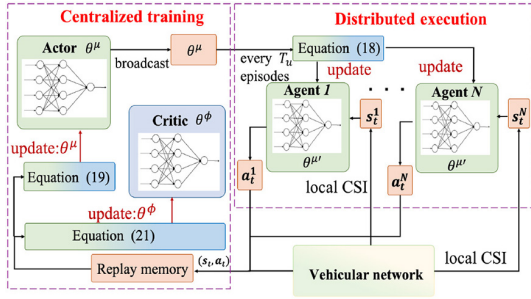


Fig. 3. Diagram of the proposed DDPG-based algorithm.

θ^ϕ and θ^ϕ_I), respectively. Then, the parameters of the target networks are softly updated with that of estimated networks as Equation (15), where $\xi \ll 1$ [14].

$$\begin{aligned}\theta^{\mu_I} &\leftarrow \xi \theta^{\mu} + (1 - \xi) \theta^{\mu_I} \\ \theta^{\phi_I} &\leftarrow \xi \theta^{\phi} + (1 - \xi) \theta^{\phi_I}.\end{aligned}\quad (15)$$

Given the experiences in $\mathcal{B}$, the actor and critic interact with each other to update θ^{μ} and θ^{ϕ} . Specifically, the actor updates θ^{μ} by making policy $\mu(\cdot)$ approximate the optimal policy π to achieve the maximum $\mathbb{E}[Q^{\pi}]$. Thus, the policy objective function, $J = \mathbb{E}_{\theta^{\mu}}[Q]$, is used to evaluate the $\mu(\cdot)$. Therefore, θ^{μ} can be updated with the gradients of J as follows:

$$\Delta \theta^{\mu} = \lambda_a \frac{1}{B} \sum_B \nabla_a Q(s_t, \mu(s_t) | \theta^{\phi}) \nabla_{\theta^{\mu}} \mu(s | \theta^{\mu}) |_{s=s_t} \quad (16)$$

where λ_a is the learning rate of the actor.

Moreover, the critic evaluates whether the action taken by actor is appropriate via the critic's action-value function. It updates θ^{ϕ} based on the loss function by minimizing the temporal difference (TD)-error, which describes the difference between $Q(s_t, \mathbf{a}_t | \theta^{\phi})$ and $Q(s_t, \mathbf{a}_t | \theta^{\phi_I})$. Therefore, the loss function can be described as:

$$L(\theta^{\phi}) = \mathbb{E}[r_t + \gamma Q(s_{t+1}, \mathbf{a}_{t+1} | \theta^{\phi_I}) - Q(s_t, \mathbf{a}_t | \theta^{\phi})]^2. \quad (17)$$

Then, θ^{ϕ} is updated with the gradient of $L(\theta^{\phi})$ as follows:

$$\Delta \theta^{\phi} = \lambda_c \frac{1}{B} \sum_B \nabla_{\theta^{\phi}} Q(s_t, \mathbf{a}_t) \quad (18)$$

where λ_c is the learning rate of the critic and $\nabla_{\theta^{\phi}} Q(s_t, \mathbf{a}_t)$ is the gradient of $L(\theta^{\phi})$ with respect to θ^{ϕ} .

C. Implementation Considerations

1) It is critical to evaluate the computation complexity of the proposed RRM scheme, which mainly depends on Hungarian algorithm and DDPG-based algorithm. Therein, the computation complexity of Hungarian algorithm is upper bounded by $O((\max\{M, N\})^3)$. Assuming that actor network and critic network of each agent contains J and L fully connected layers, the computation complexity of the training process can be described as:

$$\mathcal{O}\left((\max\{M, N\})^3 + N \times \left(\sum_{j=0}^{J-1} n_{A,j} n_{A,j+1} + \sum_{l=0}^{L-1} n_{C,l} n_{C,l+1}\right)\right) \quad (19)$$

where $n_{A,j}$ and $n_{C,l}$ represent the number of neurons in j -th actor network layer and l -th critic network layer, respectively. $n_{A,0}$ and $n_{C,0}$ equal the input size of corresponding networks.

TABLE I
COMMUNICATION COST COMPARISON

	Communication cost of BS	Communication cost of V-UE
Centralized RRM scheme	global CSI for all links and results feedback	\
Hybrid centralized-distributed RRM scheme	global CSI for V2I links and results feedback	local CSI for V2V links

TABLE II
SIMULATION PARAMETERS

Parameter	Value
Number of V2I links (V2V links) M (N)	4-28
Carrier frequency/ Bandwidth	2GHz / 4MHz
BS/Vehicle antenna gain	8dBi/3dBi
BS/Vehicle receiver noise figure	5dB/9dB
Absolute vehicle speed	36km/h-54km/h
S / P_{max}^{max}	23dBm / 23dBm
V2V SINR threshold $\gamma_{n,min}$	5dB
Noise power σ^2	-114dBm
Replay memory size D / Batch size B	10000 / 32
Learning rate of the actor/critic λ_a/λ_c	0.0001/0.001
Discount factor γ	0.97
Total episodes of training / testing	3000 / 100
Update period of target network T_u	5 episodes
Hidden layer	500,250,120 (ReLU)

2) The communication cost of system deployment is also important for the utility of schemes. As shown in Table I, compared with centralized RRM schemes, the proposed RRM scheme relieves central communication pressure by utilizing edge computation capacity of distributed V-UEs.

It is worth noting that the communication cost is caused by the network parameters of actor network. Thus, the size of that for every agent can be described as:

$$\sum_{j=0}^{J-1} n_{A,j} n_{A,j+1} \quad (20)$$

As calculated in Equation (20), the size of network parameters is approximately equal to 3.8MB. In this case, the high throughput of C-V2X, especially 5G-V2X (hundreds of Mbps), can achieve it in millisecond [15]. Moreover, the number of the vehicles served by the same BS is limited. Meanwhile, the V-UE is updated once for a longer period, like once a couple of hours. In this case, the communication cost of the interaction between the BS and agents is mild and affordable based on the current C-V2X technologies.

IV. NUMERICAL SIMULATION

In this section, we consider the urban case and set up simulation following the evaluation methodology defined by 3GPP [2]. The major parameters are listed in Table II. Here, we explore the performance including runtime, system capacity and outage probability of the proposed RRM scheme. In detail, channel allocation is performed according to the Proposition 2. Furthermore, we implement DDPG-based algorithm and compare it with other two algorithms for power control (i.e., DQN-based and random power selection algorithm (RPSA)).

From the perspective of numerical simulation, we further assess the computation complexity between different schemes. As shown in Fig. 2, the BS transforms channel allocation into an MWM problem and handles it via Hungarian algorithm in polynomial time. Accordingly, it is conducted within several milliseconds by simulation. Thereby, we focus on the runtime for power control as shown in Table III. Here, DQN-based

TABLE III
 RUNTIME COMPARISON

	Quantization level (dBm)	Training time (seconds) / Episodes	Testing time (milliseconds)
DQN-2	[23,5]	253 / 1200	450
DQN-4	[23,15,5,-100]	453 / 2000	520
DQN-8	[23,19,15,10,5,0,-50,-100]	Unavailable	Unavailable
DDPG	No quantization	530 / 6500	670

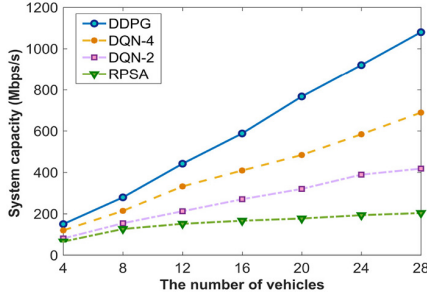


Fig. 4. System capacity comparison among different schemes.

algorithm quantizes action space and three quantization types are considered including DQN-2, DQN-4 and DQN-8 (where 2, 4, 8 denote the quantization levels). It is observed that the training time rises with the refinement of action space. Moreover, it is worth noting that DQN-8 cannot run due to the explosive space complexity. Thus, DQN-based algorithm shows limitations on large-dimension action space. In addition, the testing process of the DDPG-based algorithm is accomplished in a couple of tenths of a second. In this case, the practical implementation of our proposed scheme is able to adapt to the time-varying environment.

Fig. 4 compares the system capacity between different algorithms. It shows that the system capacity increases quasi-linearly with the number of vehicles. The reason is that every V2I link is orthogonal and allows just one V2V link to share spectrum with it. It means that the system capacity of V2I links enhances as the number of vehicles increases. Meanwhile, the system capacity of V2V links rises while occupying spectrum with V2I links. In addition, higher system capacity is achieved by the DDPG-based algorithm, which coincides with the optimal solution.² Moreover, the system capacity of that remains higher ascending rate due to accumulative performance gap. Hence, we conclude that our proposed scheme provides a tight lower bound for **RRM**. This is because that DDPG-based algorithm searches the optimal policy in global space.

In Fig. 5, we show the outage probability of V2V links, which reflects the reliability of communication links [18], [19], due to the constraint (5e) of **RRM**. The results illustrate that the outage probability of V2V links increases as the number of vehicles grows. The reason is that there are more V2V links sharing spectrum with V2I links and suffering more severe co-channel interference. The DDPG-based algorithm shows superior reliability because its reward design penalizes the policy that violates the SINR requirement of V2V links.

²The optimal solution [16] to **RRM** problem can be obtained under the specific condition [17, eq. (9c)]. When this condition is satisfied, we find that the gap between our proposed scheme and the optimal solution is zero by numerical simulation. However, when this condition is violated, our proposed scheme achieves higher system capacity than the solution in [16].

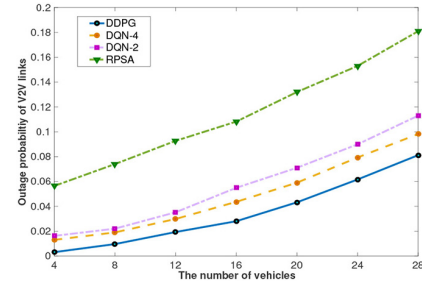


Fig. 5. Comparison of outage probability of V2V links.

Moreover, it can flexibly adjust the transmission power in the feasible region for every V-UE. Therefore, the learned policy can more strictly satisfies the SINR requirement of V2V links.

REFERENCES

- [1] K. J. Ahmed and M. J. Lee, "Secure LTE-based V2X service," *IEEE Internet Things J.*, vol. 5, no. 5, pp. 3724–3732, Oct. 2018.
- [2] *Technical Specification Group Radio Access Network: Study LTE-Based V2X Services: (Release 14), Version 2.0.0*, 3GPP Standard (TS) 36.885, Apr. 2016.
- [3] W. Sun et al., "Radio resource management for D2D-based V2V communication," *IEEE Trans. Veh. Technol.*, vol. 65, no. 8, pp. 6636–6650, Aug. 2016.
- [4] M. Gonzalez-Martín et al., "Analytical models of the performance of C-V2X mode 4 vehicular communications," *IEEE Trans. Veh. Technol.*, vol. 68, no. 2, pp. 1155–1166, Feb. 2019.
- [5] L. Liang et al., "Graph-based radio resource management for vehicular networks," in *Proc. IEEE ICC*, 2018, pp. 1–6.
- [6] J. Mei et al., "A latency and reliability guaranteed resource allocation scheme for LTE V2V communication systems," *IEEE Trans. Wireless Commun.*, vol. 17, no. 6, pp. 3850–3860, Jun. 2018.
- [7] M. K. Abdel-Aziz et al., "Ultra-reliable and low-latency vehicular communication: An active learning approach," *IEEE Commun. Lett.*, vol. 24, no. 2, pp. 367–370, Feb. 2020.
- [8] S. Maghsudi and S. Stanczak, "Hybrid centralized-distributed resource allocation for device-to-device communication underlying cellular networks," *IEEE Trans. Veh. Technol.*, vol. 65, no. 4, pp. 2481–2495, Apr. 2016.
- [9] S. Gyawali et al., "Resource allocation in vehicular communications using graph and deep reinforcement learning," in *Proc. IEEE GLOBECOM*, 2019, pp. 1–6.
- [10] C.-Y. Wei et al., "QoS-aware hybrid scheduling for geographical zone-based resource allocation in cellular vehicle-to-vehicle communications," *IEEE Commun. Lett.*, vol. 22, no. 3, pp. 610–613, Mar. 2018.
- [11] L. Liang et al., "Spectrum sharing in vehicular networks based on multi-agent reinforcement learning," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 10, pp. 2282–2292, Oct. 2019.
- [12] Y. S. Nasir and D. Guo, "Multi-agent deep reinforcement learning for dynamic power allocation in wireless networks," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 10, pp. 2239–2250, Oct. 2019.
- [13] Q. Wei et al., "Resource allocation for V2X communications: A local search based 3D matching approach," in *Proc. IEEE ICC*, 2017, pp. 1–6.
- [14] H. Peng and X. Shen, "Deep reinforcement learning based resource management for multi-access edge computing in vehicular networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 4, pp. 2416–2428, Oct.–Dec. 2020.
- [15] K. Sehla et al., "Resource allocation modes in C-V2X: From LTE-V2X to 5G-V2X," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 8291–8314, Jun. 2022.
- [16] D. Feng et al., "Device-to-device communications underlying cellular networks," *IEEE Trans. Commun.*, vol. 61, no. 8, pp. 3541–3551, Aug. 2013.
- [17] A. Gjendemsjo et al., "Optimal power allocation and scheduling for two-cell capacity maximization," in *Proc. 4th ISMOM Ad Hoc Wireless Netw.*, 2006, pp. 1–6.
- [18] Y. Yuan et al., "Meta-reinforcement learning based resource allocation for dynamic V2X communications," *IEEE Trans. Veh. Technol.*, vol. 70, no. 9, pp. 8964–8977, Sep. 2021.
- [19] J. Gao et al., "Deep neural network based resource allocation for V2X communications," in *Proc. VTC-Fall*, 2019, pp. 1–5.